

LOCALIZED BOST–CONNES SYSTEMS OVER GLOBAL FUNCTION FIELDS: COMMENSURABILITY FORBIDS TYPE III₁, AND THE FACTOR RECOVERS THE TEMPERATURE

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ABSTRACT. We transport the localized Bost–Connes theory to $K = \mathbb{F}_q(C)$ for a smooth projective geometrically irreducible curve $C/\mathbb{F}_q$ of genus g , and find that one structural feature inverts and, with it, the central negative result of Paper IX of the series.

Over a number field the logarithms $\log \mathbf{Np}$ of an infinite prime set are never pairwise commensurable [5, Lem. 4.1]; over a function field they are *always* commensurable, being $\deg(P) \log q$. Hence $\Gamma_\beta = \langle q^{-\beta \deg P} \rangle = q^{-\beta d_S \mathbb{Z}}$ with $d_S = \gcd_{P \in S} \deg P$ is a *discrete* subgroup of $\mathbb{R}_{>0}$, so $S(M) \subseteq \Gamma_\beta \cup \{0\}$ can never be $[0, \infty)$: **the extremal KMS _{β} factors are never of type III₁**, in complete contrast with [5, Thm. 3.2], where III₁ is generic. The proposed alternative “III₀ if Γ_β is dense” cannot occur, a discrete group not being dense; the alternatives are III _{λ} with $\lambda \in q^{-\beta d_S \mathbb{Z}}$, III₀, or semifiniteness.

The consequence is the interesting one. In the III _{λ} case $M_\beta \cong R_\lambda$ with $\lambda = q^{-\beta d_S}$, and the Powers factors are mutually non-isomorphic; so **the von Neumann algebra determines $\beta d_S \log q$** . Paper IX showed that over number fields every system with $S = P_K$, $\beta \leq 1$ gives R_∞ and the factor remembers neither K , nor S , nor β . Over function fields the collapse is partial: the temperature survives weak closure. This is the first repair of Theorem 2.1 of Paper IX in the whole programme, and it is bought by commensurability rather than by any rigidity input.

Two corrections. The modular flow is periodic with fundamental period $T_0 = 2\pi/(d_S \log q)$, not $2\pi/\log q$; the two agree exactly when $d_S = 1$, which does hold for $S = \mathcal{P}_K$ by F. K. Schmidt’s theorem but fails for, say, the set of points of even degree. And in the Weil–Chebotarev bound the term $(g_L + 1)q^{n/2}$ swamps the main term $2g_L q^{n/2}/n$ and is superfluous: the truth is $O_\chi(q^{n/2}/n)$, with the constant-field-extension characters requiring a separate and elementary treatment, Weil’s bound not applying to them.

The rest goes through and is unconditional. $\beta_c = 1$ for every curve; uniqueness holds on $(0, 1]$ for $S = \mathcal{P}_K$, unconditionally — the class-group characters, which over a number field can obstruct permanently, here have detecting sets of positive density; and a set with $A_n(S) \asymp q^n/n^2$ has $\sum_{P \in S} q^{-\deg P} = \zeta(2)$ -like and finite, giving a *closed* low-temperature phase $[1, \infty)$ with no sieve hypothesis whatever.

1. THE SYSTEM AND ITS PERIODIC FLOW

Let $C/\mathbb{F}_q$ be a smooth projective geometrically irreducible curve of genus g , $K = \mathbb{F}_q(C)$, and $\mathcal{P}_K$ the set of closed points. For $S \subseteq \mathcal{P}_K$ put

$$J_S = \operatorname{Div}_+^S(C) = \bigoplus_{P \in S} \mathbb{N} \cdot P, \quad \mathbf{N}(P) = q^{\deg P}, \quad \mathcal{A}_{C,S} = C(Y_S) \rtimes J_S,$$

with $G_S = \mathbb{I}_S / \overline{K_S^\times}$ and $\sigma_t(\mu_D) = q^{it \deg D} \mu_D$. Write

$$d_S := \gcd_{P \in S} \deg P.$$

Sources, code and data for the entire series: <https://github.com/Ruqing1963/localized-bost-connes>; this paper is in [papers/XVII-function-fields/](#).

Constructions and notation follow [4]; Papers VIII, IX, XIII and XV of the series, referred to by numeral, are unpublished companions, and every fact used from them is restated or verified in place. One convention is needed at once: for $S = \mathcal{P}_K$ the degree map gives $\mathbb{I}_S/\overline{K_S^\times}$ a quotient $\mathbb{Z}$, so the group as written is not compact; as the framework of [4] requires a compact symmetry group (and as is standard for Bost–Connes systems over function fields), G_S denotes throughout the profinite completion in the degree direction. Its dual $\widehat{G}_S$ is then a torsion group, and the characters factoring through the constant-field tower $\text{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q) \cong \widehat{\mathbb{Z}}$ are exactly $P \mapsto \zeta^{\deg P}$ with ζ a root of unity. For the basic theory of curves over finite fields we refer to [1, 2].

Proposition 1.1 (Periodicity). *The flow σ is periodic with fundamental period*

$$T_0 = \frac{2\pi}{d_S \log q}.$$

In particular $T_0 = 2\pi/\log q$ precisely when $d_S = 1$, which holds for $S = \mathcal{P}_K$ by F. K. Schmidt’s theorem [1] that the gcd of the degrees of closed points of a curve over a finite field is 1; it fails for instance when S consists of the points of even degree, where $d_S = 2$ and the period is halved.

Proof. $\sigma_T = \text{id}$ iff $q^{iT \deg D} = 1$ for every $D \in J_S$, i.e. $T \log q \deg D \in 2\pi\mathbb{Z}$. The degrees realized by J_S generate the subgroup $d_S\mathbb{Z}$ of $\mathbb{Z}$, so the condition is $T d_S \log q \in 2\pi\mathbb{Z}$, whose least positive solution is T_0 . $\square$

Remark 1.2. Periodicity of σ has no analogue over number fields, where the $\log N\mathfrak{p}$ generate a dense subgroup of $\mathbb{R}$ and σ is never periodic. It means that σ factors through the circle $\mathbb{R}/T_0\mathbb{Z}$, so $\mathcal{A}_{C,S}$ carries a genuine $\mathbb{T}$ -action rather than a flow.

2. RATIONAL THERMODYNAMICS

Proposition 2.1 (Weil zeta). *With $T = q^{-\beta}$,*

$$\prod_{P \in S} (1 - q^{-\beta \deg P})^{-1} = Z_S(C, T),$$

and for $S = \mathcal{P}_K$ this is the Weil zeta function [3] $Z_C(T) = \frac{P(T)}{(1-T)(1-qT)}$ with $P(T) = \prod_{j=1}^{2g} (1 - \alpha_j T)$ and $|\alpha_j| = q^{1/2}$. Its poles are at $T = 1$ and $T = q^{-1}$, that is at $\beta = 0$ and $\beta = 1$.

Corollary 2.2 (Critical exponent). *For every curve C and $S = \mathcal{P}_K$, $\beta_c(C) = 1$: the series $\sum_P q^{-\beta \deg P} = \sum_n a_n q^{-\beta n}$, with $a_n = \#\{P : \deg P = n\} = \frac{q^n}{n} + O(\frac{q^{n/2}}{n})$ by Weil, converges for $\beta > 1$ and diverges for $\beta \leq 1$.*

Proof. $\sum_n \frac{q^{n(1-\beta)}}{n}$ diverges for $\beta \leq 1$, converging otherwise; the Weil error term contributes $\sum_n q^{n(1/2-\beta)}/n$, convergent for $\beta > 1/2$. $\square$

Remark 2.3 (Numerical illustration). Partial sums $\sum_{n \leq 40} a_n q^{-\beta n}$ with $a_n = q^n/n$: for $q = 2$ they are 11.36, 4.28, 2.69 at $\beta = 0.9, 1.0, 1.1$, and for $q = 4$ they are 65.68, 4.28, 2.04; the value at $\beta = 1$ is $\sum_{n \leq 40} 1/n$ in both cases, growing logarithmically, and the $\beta = 1.1$ column is bounded.

3. COMMENSURABILITY FORBIDS TYPE III₁

Lemma 3.1 (Total commensurability). *For every $P \in S$, $\log N(P) = \deg(P) \log q \in (\log q)\mathbb{Z}$. Hence the log-norms are pairwise commensurable, and*

$$\Gamma_\beta = \langle N(P)^{-\beta} : P \in S \rangle = q^{-\beta d_S \mathbb{Z}},$$

a discrete subgroup of $(\mathbb{R}_{>0}, \times)$.

Remark 3.2 (The inversion). [5, Lem. 4.1] proves that over a number field an infinite S never has pairwise commensurable log-norms, distinct rational primes having $\mathbb{Q}$ -independent logarithms. Over a function field the opposite holds universally. This single reversal drives everything below.

Theorem 3.3 (Never III_1). *For every curve C , every S and every $\beta > 0$, the extremal KMS_β factors $M = \pi_\varphi(\mathcal{A}_{C,S})''$ are not of type III_1 . Moreover Γ_β being discrete, the case “ Γ_β dense” cannot occur, so the possibilities are: type III_λ with $\lambda \in q^{-\beta d_S \mathbb{Z}}$; type III_0 , where $S(M) = \{0, 1\}$; or semifinite, where $S(M) = \{1\}$.*

Proof. By [4, Prop. 6.1], $S(M) \subseteq \overline{\Gamma_\beta} \cup \{0\}$, and $\overline{\Gamma_\beta} = \Gamma_\beta$ by Lemma 3.1. Type III_1 requires $S(M) = [0, \infty)$, which is not contained in a discrete set together with 0. The listed cases exhaust Connes’ classification of $S(M)$. $\square$

Corollary 3.4 (Contrast). *By [5, Thm. 3.2], over a number field with $S = P_K$ and $0 < \beta \leq 1$ the type is always III_1 . Over a function field it never is, for any S and any β . The two settings are opposite in this respect, and the reason is Lemma 3.1 alone, no arithmetic being used.*

4. THE FACTOR RECOVERS THE TEMPERATURE

Theorem 4.1 (Partial repair of Paper IX). *Suppose $M_\beta = \pi_{\varphi_\beta}(\mathcal{A}_{C,S})''$ is of type III_λ with $\lambda = q^{-\beta d_S}$. Then $M_\beta \cong R_\lambda$, the Powers factor: $\mathcal{A}_{C,S}$ is amenable — the argument of Paper IX (its Lemma 1.1) applies verbatim, the acting group $I_S = \mathbb{Z}^{(S)}$ being abelian here too, so the groupoid is amenable and M_β is injective — and since $R_\lambda \cong R_{\lambda'}$ only for $\lambda = \lambda'$, the isomorphism class of M_β determines*

$$\lambda = q^{-\beta d_S}, \quad \text{hence} \quad \beta d_S \log q.$$

In particular, for fixed q and d_S , distinct temperatures give non-isomorphic factors.

Proof. Amenability and Connes’ classification give $M_\beta \cong R_\lambda$; the Powers factors are pairwise non-isomorphic, their S -invariants $\{0\} \cup \lambda \mathbb{Z}$ being distinct. $\square$

Remark 4.2 (What this changes). Theorem 2.1 of Paper IX showed that over number fields $M_{\mathbb{Q}, P_{\mathbb{Q}}, 1} \cong M_{\mathbb{Q}, P_{\mathbb{Q}}, 1/2} \cong M_{K, P_K, \beta} \cong R_\infty$ for every K and every $\beta \in (0, 1]$: the factor remembers neither the field, nor the prime set, nor the temperature. Theorem 4.1 is the first place in this programme where that collapse is not total. It is worth being precise about how much is repaired: the factor recovers the single real number $\beta d_S \log q$, and nothing else — not C , not g , not S beyond d_S , not the obstruction group. But β was the quantity whose loss was most striking in Paper IX, and it is recovered here for free, by commensurability rather than by any rigidity theorem.

Numerically:

(q, d_S, β)	$(2, 1, \frac{1}{2})$	$(2, 1, 1)$	$(2, 1, 2)$	$(4, 2, 1)$	$(9, 1, \frac{3}{2})$
λ	0.7071	0.5000	0.2500	0.0625	0.0370

Distinct temperatures give distinct factors.

Remark 4.3 (The remaining ambiguity). Only the product $\beta d_S \log q$ is recovered, so (q, d_S, β) and (q', d'_S, β') with $\beta d_S \log q = \beta' d'_S \log q'$ give isomorphic factors: for example $(q, d_S, \beta) = (4, 1, 1)$ and $(2, 1, 2)$ both give $\lambda = 1/4$. Recovering q separately requires the C^* -level, where by Proposition 1.1 of Paper XV the point spectrum $\text{Sp}_p(\sigma) = q^{d_S \mathbb{Z}}$ is available and pins q^{d_S} directly.

5. THE WEIL–CHEBOTAREV OBSTRUCTION

Proposition 5.1 (Corrected bound). *Let χ be a nontrivial character of G_S that does not factor through the constant-field tower; write $\chi = z^{\deg} \cdot \chi_0$ with z a root of unity and χ_0 geometric with*

conductor $\mathfrak{f}_\chi$, and put $B_\chi = 2g - 2 + \deg \mathfrak{f}_\chi$ [2]. Then $L(T, \chi) = L(zT, \chi_0) = \prod_{j=1}^{B_\chi} (1 - \alpha_j T)$ with $|\alpha_j| = q^{1/2}$ by Weil [3], and

$$\left| \sum_{\deg P=n} \chi(\text{Frob}_P) \right| \leq \frac{B_\chi}{n} q^{n/2} + \frac{4g+4}{n} q^{n/2} \quad (n \geq 1).$$

Proof. Taking logarithmic derivatives of $L(T, \chi)$ gives $\sum_{\deg P|n} \deg P \cdot \chi(\text{Frob}_P)^{n/\deg P} = -\sum_j \alpha_j^n$, of modulus at most $B_\chi q^{n/2}$. The terms with $\deg P = m < n$ are bounded in absolute value by $m a_m \leq \#C(\mathbb{F}_{q^m}) \leq q^m + 1 + 2gq^{m/2} \leq (2g+2)q^m$, and $\sum_{m \leq n/2} (2g+2)q^m \leq (2g+2) \frac{q}{q-1} q^{n/2} \leq (4g+4)q^{n/2}$; removing them and dividing by n gives the statement. $\square$

Remark 5.2 (The proposed bound). The bound $2g_L q^{n/2}/n + (g_L + 1)q^{n/2}$ is true but its second term has no factor $1/n$ and therefore dominates the first for every $n \geq 1$, making the whole estimate weaker than the trivial $O(q^{n/2})$ that the main term already provides. The correct order is $O_\chi(q^{n/2}/n)$, as in Proposition 5.1.

Remark 5.3 (Constant field extensions need separate treatment). Weil's bound does not apply to the characters χ factoring through $\text{Gal}(\mathbb{F}_{q^m}/\mathbb{F}_q)$, for which $\chi(\text{Frob}_P) = \zeta^{\deg P}$ with ζ a primitive m -th root of unity: the corresponding L -function is $Z_C(\zeta T)$, which has poles rather than the required polynomial form. These are the function field counterpart of the class-group characters of [4, §4.1], and they must be handled directly. That is elementary: $D_\chi = \{P : \deg P \not\equiv 0 \pmod{m}\}$, and $\sum_{P \in D_\chi} q^{-\beta \deg P} = \sum_{n \not\equiv 0 \pmod{m}} a_n q^{-\beta n}$ diverges for $\beta \leq 1$, since $a_n \sim q^n/n$ on the nonzero residue classes as well (Corollary 2.2).

Theorem 5.4 (Unconditional uniqueness). *Let $S = \mathcal{P}_K$. For every nontrivial $\chi \in \widehat{G_S}$,*

$$\sum_P q^{-\beta \deg P} |1 - \chi(\sigma_P)|^2 = \infty \quad (0 < \beta \leq 1),$$

so $\Xi_\beta = \{1\}$ and the KMS_β state is unique for $0 < \beta \leq 1$, unconditionally and with explicit constants. In particular the unramified class-group obstruction of [4, §4.1] has no analogue at $S = \mathcal{P}_K$: the characters pulled back from $\text{Pic}(C)$ have detecting sets of positive density.

Proof. Since $|1 - w|^2 = 2 - 2 \text{Re } w$ on the unit circle,

$$\sum_{\deg P=n} |1 - \chi(\text{Frob}_P)|^2 = 2a_n - 2 \text{Re} \sum_{\deg P=n} \chi(\text{Frob}_P).$$

If χ does not factor through the constant-field tower, Proposition 5.1 bounds the second term by $2(B_\chi + 4g + 4)q^{n/2}/n$, so the sum is $\geq a_n$ for n large and $\sum_n q^{-\beta n} a_n = \infty$ for $\beta \leq 1$ by Corollary 2.2. If $\chi = \zeta^{\deg}$ is a constant-field character, Remark 5.3 applies. These two cases are exhaustive, $\widehat{G_S}$ being torsion. Hence $\chi \notin \Xi_\beta$ for every nontrivial χ and $0 < \beta \leq 1$, and uniqueness follows from the criterion of [4, Thm. 3.9], whose Kakutani–martingale proof uses only the product structure of the measures and applies verbatim over a function field. For the last sentence: a character of $\text{Pic}(C)$ pulled back to G_S is $\zeta^{\deg}$ times a character of the finite group $\text{Pic}^0(C)$; the latter is geometric and everywhere unramified, with $B_\chi = 2g - 2$, so it falls under the first case, and its detecting set has density bounded below explicitly in g . $\square$

Remark 5.5 (What is gained over the number field case). The corresponding statement over $\mathbb{Q}$ [4, Thm. 3.9] relies on the prime ideal theorem and Chebotarev density with their ineffective error terms, Siegel–Walfisz and the possibility of Landau–Siegel zeros. Here Weil's theorem is available and every constant is explicit in g and $\deg \mathfrak{f}_\chi$. Nothing is conjectural.

6. UNCONDITIONAL SPARSE SETS AND CLOSED PHASES

Theorem 6.1 (Closed low-temperature phase, no sieve). *Choose, for each n , any $A_n = \lfloor q^n/n^2 \rfloor$ closed points of degree n ; this is possible for n large since $a_n \sim q^n/n \gg q^n/n^2$. Let S_{geom} be their union. Then*

$$\sum_{P \in S_{\text{geom}}} q^{-\beta \deg P} \asymp \sum_n \frac{q^{n(1-\beta)}}{n^2},$$

so $\beta_c(S_{\text{geom}}) = 1$ and the series converges at $\beta = 1$, with value $\asymp \zeta(2) = \pi^2/6$. Consequently the symmetry-broken phase is the closed half-line $[1, \infty)$, exactly as in [4, Thm. 5.5], and the construction uses no sieve hypothesis of any kind.

Proof. The displayed asymptotic is immediate; $\sum_n n^{-2} < \infty$ gives convergence at $\beta = 1$ and $\sum_n q^{n(1-\beta)}/n^2 = \infty$ for $\beta < 1$ gives $\beta_c = 1$. Closedness of the phase is [4, Thm. 5.5], whose hypothesis is exactly convergence at the critical exponent. $\square$

Remark 6.2 (Numerical check). With $q = 2$, $\sum_{n \leq 200} 2^{n(1-\beta)}/n^2$ equals 2.964, 1.640, 1.493 at $\beta = 0.95, 1.00, 1.05$; the middle value is the truncation of $\pi^2/6 = 1.6449$.

Remark 6.3 (Comparison). Over $\mathbb{Q}$ the closed phase of [4, Thm. 5.5] was obtained for the Sophie Germain primes and required Brun's sieve; the corresponding statement for the twin primes remains tied to the Hardy–Littlewood conjecture. Over a function field the point counts are known exactly, so a set of any prescribed density can simply be selected. The gain is not that a hard theorem becomes easy but that the question stops being about existence: one chooses the set rather than proving that one exists.

7. DICTIONARY AND ASSESSMENT

Number field K	Function field $\mathbb{F}_q(C)$	Reference
$\log N_{\mathfrak{p}}$ never commensurable	always commensurable	[5, Lem. 4.1], Lem. 3.1
σ never periodic	periodic, $T_0 = 2\pi/(d_S \log q)$	Prop. 1.1
β_c depends on S	$\beta_c = 1$ for $S = \mathcal{P}_K$	Cor. 2.2
type III ₁ generic	never III ₁	Thm. 3.3
$M \cong R_\infty$, β forgotten	$M \cong R_\lambda$ recovers $\beta d_S \log q$	Thm. 4.1
Chebotarev ineffective	Weil bound, explicit	Prop. 5.1
sparse sets need sieves	sets are chosen, not found	Thm. 6.1
unramified obstruction ([4, §4.1])	none at $S = \mathcal{P}_K$: positive density	Thm. 5.4

The transfer succeeds and, unlike the transfer to arithmetic topology of Paper VIII, it succeeds with content: every analytic difficulty of the number field theory becomes a theorem, and one structural feature genuinely inverts. That inversion, total commensurability of the log-norms, is the whole story. It makes the flow periodic, it makes Γ_β discrete, it forbids type III₁, and it thereby causes the von Neumann algebra to retain the temperature that Paper IX showed is destroyed over number fields.

We should be exact about the size of that last claim. The factor recovers one real number, $\beta d_S \log q$, and by Remark 4.3 not even q and β separately. It does not recover the curve, its genus, or the obstruction group; the collapse of Paper IX is weakened, not reversed. But it is the first weakening, and it comes from the arithmetic of the ground field rather than from any additional structure imposed on the system.

Three questions. Which of the three cases of Theorem 3.3 actually occurs, and for which S ? The natural guess is III _{$q^{-\beta d_S}$} throughout, but determining Σ_β exactly, as Paper I does over number fields ([5, Prop. 2.2] for the bound via the extended group, [5, Thm. 4.2] for an exact

determination), requires computing the extended obstruction group here and we have not done it. Second, does the periodicity of σ — so that $\mathcal{A}_{C,S}$ carries a circle action rather than a flow — bring the classification of Paper XIII within reach of the equivariant Kirchberg theory for compact groups, which is far better developed than for $\mathbb{R}$ (its Remark 5.1)? Third, the Weil bound gives effective control on Ξ_β that has no number field counterpart; whether it yields a realization theorem for prescribed chains $\{\Xi_\beta\}$, the function field analogue of [5, Thm. 5.1], is open and looks tractable.

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